

Ex. No. : 01A

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GUI Components

Aim:

Develop an application to change the font and color of the text and display toast message when the user presses the button.

Procedure:

1. Create a New Android Project
 - Open Android Studio.
 - Click on New Project → Select Empty Activity.
 - Enter the Application Name, Package Name, and select Kotlin as the language.
 - Click Finish.
2. Configure AndroidManifest.xml
 - Open AndroidManifest.xml.
 - Ensure that the MainActivity is declared.
 - Set the launcher intent so that the app starts with MainActivity.
3. Design the User Interface
 - Open activity_main.xml.
 - Use a LinearLayout with vertical orientation.
 - Add a TextView to display text.
 - Add three Buttons:
 - Change Font Size
 - Change Font Color
 - Change Background Color
 - Assign proper IDs to all views.
4. Write Kotlin Code
 - Open MainActivity.kt.
 - Use setContentView() to load the layout.

- Access UI elements using findViewById().
 - Implement setOnClickListener() for each button:
 - Font Size Button: Increases text size cyclically.
 - Font Color Button: Changes text color among predefined colors.
 - Background Color Button: Changes background color of the layout.
5. Run the Application
- Connect an Android Emulator or physical device.
 - Click Run ► in Android Studio.
 - Observe the changes when buttons are clicked.
6. Verify Output
- Text size changes on clicking Change Font Size.
 - Text color changes on clicking Change Font Color.
 - Background color changes on clicking Change BG Color.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Experiment1_MADD_46">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
```

```

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    >

    <TextView
        android:id="@+id/tv"
        android:gravity="center"
        android:layout_width="match_parent"
        android:layout_height="232dp"
        android:text="HII"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.273" />

    <Button
        android:id="@+id/btntxtsize"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Change Font Size"
        tools:layout_editor_absoluteX="149dp"
        tools:layout_editor_absoluteY="257dp" />

    <Button
        android:id="@+id/btntxtcolor"

```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Change Font Color"
        tools:layout_editor_absoluteX="129dp"
        tools:layout_editor_absoluteY="321dp" />

<Button
    android:id="@+id/bgcolor"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Change BG Color"
    tools:layout_editor_absoluteX="149dp"
    tools:layout_editor_absoluteY="394dp" />

</LinearLayout>

```

MainActivity.kt

```

package com.example.experiment_1_madd_46

import android.os.Bundle
import android.widget.TextView
import android.widget.Button
import android.graphics.Color
import android.widget.LinearLayout
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)

        val main = findViewById<LinearLayout>(R.id.main)
        val tv = findViewById<TextView>(R.id.tv)
        val btntxtsize = findViewById<Button>(R.id.btntxtsize)
        val btntxtcolor = findViewById<Button>(R.id.btntxtcolor)
        val bgcolor = findViewById<Button>(R.id.bgcolor)
    }
}

```

```

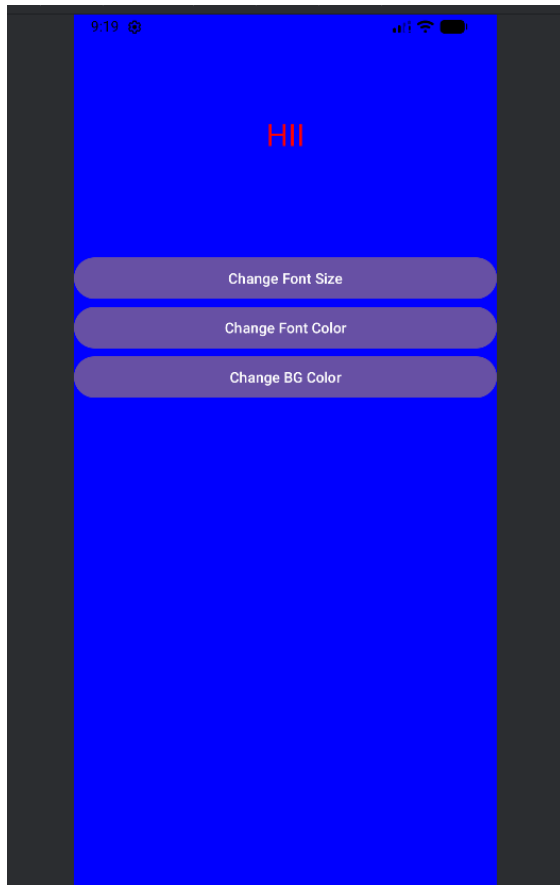
var fontSize: Float = 10f
btntxtsize.setOnClickListener {
    tv.setTextSize(fontSize)
    fontSize = ((fontSize+10)%50)
}

var fontColor: Int=5
btntxtcolor.setOnClickListener {
    when(fontColor%5){
        0 -> tv.setTextColor(Color.RED)
        1 -> tv.setTextColor(Color.YELLOW)
        2 -> tv.setTextColor(Color.BLUE)
        3 -> tv.setTextColor(Color.GREEN)
        4 -> tv.setTextColor(Color.MAGENTA)
    }
    fontColor++
}

var bg: Int=5
bgcolor.setOnClickListener {
    when(bg%5){
        1 -> main.setBackgroundColor(Color.RED)
        2 -> main.setBackgroundColor(Color.YELLOW)
        3 -> main.setBackgroundColor(Color.BLUE)
        4 -> main.setBackgroundColor(Color.GREEN)
        0 -> main.setBackgroundColor(Color.MAGENTA)
    }
    bg++
}
}
}

```

Output:



Result:

The Android application was successfully developed to dynamically change text size, text color, and background color using button click events.